PCCST 501 COMPUTER NETWORKS

Prepared by,
Ms.Neethu Krishna,
Assistant Professor ,
Department of Computer Science and Engineering,
SCMS School of Engineering & Technology,Ernakulam



SCMS SCHOOL OF ENGINEERING & TECHNOLOGY
Vidya Nagar, Palissery, Karukutty, Ernakulam - 683576,Kerala, India.
www.scmsgroup.org/sset

SEMESTER S5

COMPUTER NETWORKS

(Common to CS/CD/CM/CR/CA/AD/AI/CB/CN/CU/CI)

Course Code	**PCCST501**	**CIE Marks**	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	**ESE Marks**	60
Credits	4	**Exam Hours**	2 Hrs. 30 Min.
Prerequisites (if any)	None	**Course Type**	Theory

Course Objectives:

1. To introduce the core concepts of computer networking.
2. To develop a big picture of the internetworking implementation on Linux-based systems.
3. To impart an overview of network management concepts.

MODULE 4

4	SNMP, ASN.1 (Book 1 Ch 9) Physical Layer: Data and signals, Digital transmission, Analog transmission, Bandwidth utilization, Transmission media (Book 1 Ch 7)	**9**

SNMP – Simple Network Management Protocol

SNMP (Simple Network Management Protocol) is an **application-layer protocol** used for managing and monitoring devices in a network based on the **TCP/IP protocol suite**.

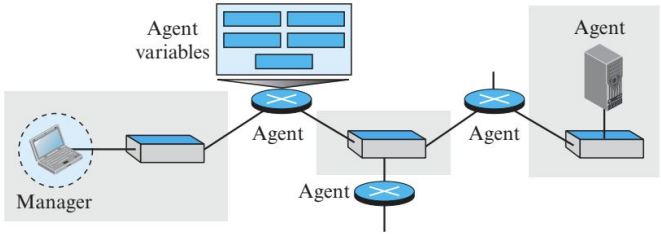
SNMP provides basic operations for **monitoring, controlling, and maintaining network devices** such as **routers and servers**.

Why is SNMP used?

SNMP is designed at the **application level**, so it can manage devices:

- From **different manufacturers**
- Connected to **different types of physical networks**
- In heterogeneous networks containing different **LANs and WANs**
- Through routers and other devices from different manufacturers

Figure 9.2 *SNMP concept*



Thus, SNMP separates network management from the specific hardware and underlying networking technology.

1. Manager and Agent

SNMP uses two main components:

Manager

A **manager** is usually a host or management station that runs the **SNMP client program**.

The manager:

- Monitors network devices
- Requests information from agents
- Can change certain values in an agent
- Can make an agent perform specific actions

Agent

An **agent** is usually a router, server, or host that runs the **SNMP server program**.

The agent:

- Collects information about the device
- Stores performance information in a **database**
- Responds to requests from the manager
- Can notify the manager when an unusual situation occurs

Manager → Monitors and controls → Agent **Agent → Provides information and warnings → Manager**

2. How Manager and Agent Work

The agent maintains information about the performance and status of the device in its database.For example, a router can store:

- Number of packets received
- Number of packets forwarded
- Other performance-related information

The manager can **request these values** from the agent and analyze them.

Example: Detecting congestion Suppose a router stores:

- Packets received = 10,000
- Packets forwarded = 6,000

The manager can compare these values to determine whether the router may be experiencing **congestion or performance problems**.

3. Manager Can Control the Agent

The manager can also make the agent perform certain actions by changing values stored in the agent's database.**Example: Remote Reboot**

A router may have a **reboot counter**.

- If the counter becomes **0**, the router reboots.
- The manager can remotely set the counter to **0**.
- The router then performs the reboot.

Therefore, SNMP allows the manager to **remotely control certain functions** of a network device.

4. Agent Can Warn the Manager – Trap

The agent does not always have to wait for the manager to ask for information.The agent continuously monitors its environment. If it detects an **unusual or abnormal condition**, it can automatically send a warning message called a **trap** to the manager.

Example

If a router detects an abnormal condition such as a serious network problem, the agent can immediately send a **trap message** to the manager.

Agent → Trap → Manager

5. Three Basic Ideas of SNMP

SNMP management is based on **three main operations**:

1. Manager checks the Agent

The manager **requests information** from the agent to monitor its behavior and performance.

Manager → Request information → Agent

2. Manager controls the Agent

The manager can **change values in the agent's database** to make the agent perform a particular task.

Manager → Change value → Agent performs task

3. Agent warns the Manager

The agent can detect an **unusual situation** and automatically send a **trap (warning message)** to the manager.

Agent → Trap → Manager

SNMP Management Components

Network management using SNMP is based on **three components**:

1. **SNMP – Simple Network Management Protocol**
2. **SMI – Structure of Management Information**
3. **MIB – Management Information Base**

1. Role of SNMP

SNMP defines **how the manager and agent communicate**.

- Defines the format of packets exchanged.
- Carries **object names and their values/status**.
- Allows the manager to **read and change values**.
- Helps interpret the received information and generate statistics.

SNMP = Communication and Management

2. Role of SMI

SMI defines the rules for describing management objects.

It specifies:

- How objects are **named**
- How objects are **organized hierarchically**
- What **data types** can be used
- How data should be represented so that different devices can understand it

Important: SMI defines the **rules**, but it does not specify which objects are managed.

SMI = Rules for naming and defining objects

3. Role of MIB

MIB defines the actual objects that are managed for a particular device.

It:

- Lists the managed objects.
- Gives each object a name according to **SMI rules**.
- Associates a **data type** with each object.
- Works like a **database structure** containing object names and types.

MIB = Actual managed objects

ASN.1 – Abstract Syntax Notation One

ASN.1 is a language used to **define the format and structure of data** exchanged between different systems.

It separates the **definition of data** from the actual **bits transmitted over the network**.

Why ASN.1 is needed?

Suppose we send **a name and a number**. The receiver must know:

- Which bits represent the name?
- Which bits represent the number?
- How are complex data such as arrays or records organized?

ASN.1 provides standard rules so that different systems can understand the same data.

Table 9.5 *Symbols used in ASN.1*

Symbol	*Meaning*	*Symbol*	*Meaning*
::=	Defined as or assignment	..	Range
|	Or, alternative, or option	{}	Start and end of a list
–	Negative sign	[]	Start and end of tag
––	The following is a comment	()	Start and end of a subtype

1. Data Types

ASN.1 provides **built-in data types**, such as:

- **INTEGER** – whole numbers
- **REAL** – real numbers
- **BOOLEAN** – TRUE/FALSE
- **ENUMERATED** – a fixed list of choices
- **SEQUENCE** – ordered list of values
- **SET** – unordered list of values
- **VisibleString** – printable characters

2. Creating New Data Types

We can create new types from existing types.

Age ::= INTEGER

MaritalStatus ::= ENUMERATED
{single, married, widowed, divorced}

DayOfWeek ::= ENUMERATED
{sun, mon, tue, wed, thu, fri, sat}

Important: A type name starts with a **capital letter**.

3. Creating Subtypes

ASN.1 allows us to restrict the range of an existing type.

```
NumberOfStudents ::= INTEGER (15..40)

Grade ::= REAL (1.0..5.0)

Weekend ::= DayOfWeek (sun | sat)
```

Here:

- NumberOfStudents → 15 to 40
- Grade → 1.0 to 5.0
- Weekend → only Sunday or Saturday

4. Variables

In ASN.1, a variable is defined with a **type and value**.

```
numberOfComputers INTEGER ::= 2

herAge INTEGER ::= 35

classSize NumberOfStudents ::= 22
```

Important: Variable names start with a **lowercase letter**.

5. Structured Data – SEQUENCE

SEQUENCE is used to create a structured data type, similar to a **struct/record in C or C++**.

Example:

```
StudentAccount ::= SEQUENCE
{
    userName VisibleString,
    password VisibleString,
    accountNumber INTEGER
}
```

A value can then be assigned:

```
johnNewton StudentAccount
{
    userName "JohnN",
    password "120007",
    accountNumber 25579
}
```

6. SEQUENCE OF

SEQUENCE OF is used when we need a collection of **similar items**, similar to an **array** in C/C++.

For example, a router's forwarding table can be represented as:

```
SEQUENCE OF Rows
```

where each **Row** can itself contain several variables.

Physical Layer

The **Physical Layer** is responsible for **transmitting bits** from the Data-Link Layer.

Main function:
Bits → Electromagnetic Signals → Transmission Medium

It deals with:

- Digital and analog signals
- Digital and analog transmission
- Data transmission speed and channel capacity
- Multiplexing
- Spread spectrum

Transmission Media

Transmission media carry signals from sender to receiver.

1. Guided Media: Uses physical cables
→ Twisted pair, Coaxial cable, Fiber optic

2. Unguided Media: Uses air/space
→ Radio waves, Microwaves, Infrared

Data and Signals

At the **Physical Layer**, communication takes place **node-to-node** using **electromagnetic signals**.

Before transmitting data, we need to understand **data and signals**.

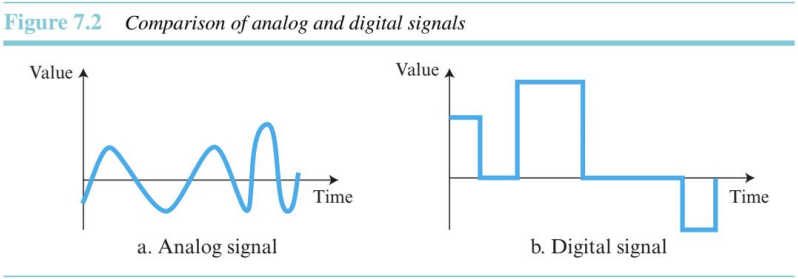
1. Analog and Digital Data

Analog Data

- **Continuous** in nature.
- Can have **infinite values**.
- Example: **Human voice, temperature, sound**.

Digital Data

- **Discrete** in nature.
- Usually represented using **0 and 1**.
- Example: **Computer data**.



2. Analog and Digital Signals

Analog Signal

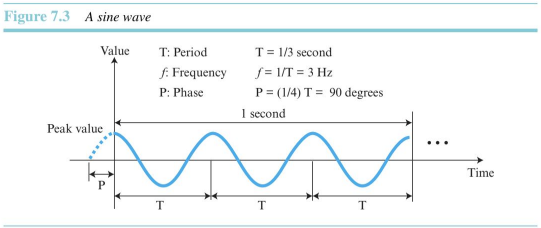
- Has **continuous values**.
- Can have **infinitely many intensity levels**.
- Example: Smooth wave.

Digital Signal

- Has a **limited number of defined levels**.
- Commonly uses **0 and 1**.

Simple idea:

Analog → Continuous
Digital → Discrete



3. Analog Signals

An analog signal can be:

Periodic Signal

A signal that **repeats its pattern** regularly.

Non-periodic Signal

A signal that **does not repeat a pattern**.

The most basic periodic analog signal is the **sine wave**.

Sine Wave – 3 Important Parameters

1. **Amplitude** – Strength/height of the signal
2. **Frequency (f)** – Number of cycles per second, measured in **Hz**
3. **Phase** – Position of the wave with respect to time

Period

Period (T) = Time required to complete **one cycle**

Relationship:

f = 1/T

4. Wavelength

Wavelength (λ) is the **distance travelled by a signal during one period**.

It depends on:

- Frequency
- Propagation speed of the medium

5. Composite Signal

A **composite signal** is made up of **multiple sine waves** having different frequencies, amplitudes, and phases.

According to **Fourier analysis**, a complex signal can be represented as a combination of simple sine waves.

6. Bandwidth

Bandwidth is the **range of frequencies contained in a signal**.

Bandwidth = Highest frequency − Lowest frequency

Example:

If frequencies range from **1000 Hz to 5000 Hz**:

Bandwidth = 5000 − 1000 = 4000 Hz

7. Digital Signals

Digital signals represent information using different signal levels.

For example:

- **1 → Positive voltage**
- **0 → Zero voltage**

A digital signal can have **2, 4, or more levels**.

If a signal has **L levels**:

Bits per level = $\log_2 L$

Signal Levels	Bits per Level
2	1 bit
4	2 bits
8	3 bits
16	4 bits

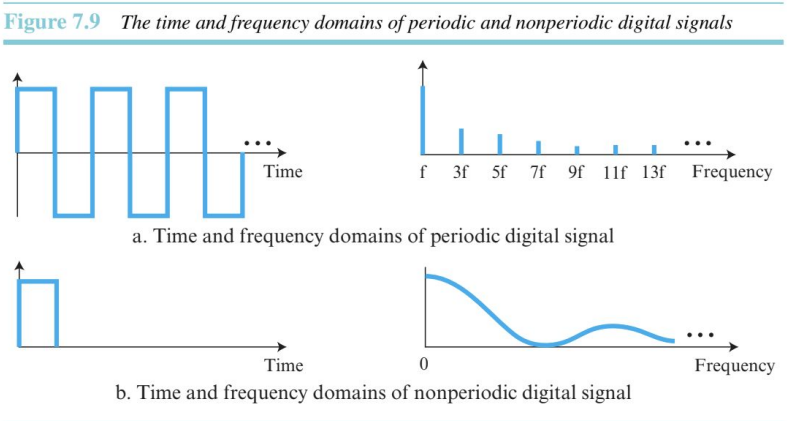


Figure 7.9 *The time and frequency domains of periodic and nonperiodic digital signals*

a. Time and frequency domains of periodic digital signal

b. Time and frequency domains of nonperiodic digital signal

8. Bit Rate

For digital signals, we use **bit rate** instead of frequency.

Bit Rate = Number of bits transmitted per second

Unit: **bps (bits per second)**

Example:

1000 bits transmitted in 1 second:

Bit rate = 1000 bps

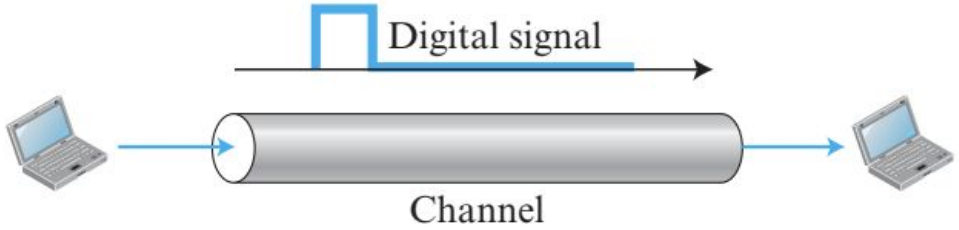
9. Bit Length

Bit length is the **distance occupied by one bit on the transmission medium**.

Digital Signal as a Composite Analog Signal

- According to **Fourier analysis**, a digital signal can be considered a **composite analog signal** made up of many frequency components. Therefore, theoretically, a digital signal has an **infinite bandwidth**. A digital signal in the time domain consists of horizontal and vertical portions. A horizontal portion represents no change in the signal, corresponding to a frequency close to **zero**, while a sudden vertical change represents very high-frequency components. Therefore, a digital signal contains a range of frequencies.
- Using **Fourier analysis**, a digital signal can be separated into its individual frequency components. If the digital signal is **periodic**, its frequency components are **discrete**, but its bandwidth is still theoretically infinite. If the digital signal is **nonperiodic**, its frequency components are **continuous**, and its bandwidth is also theoretically infinite. In data communication, digital signals are usually nonperiodic.

Figure 7.10 *Baseband transmission*



Transmission of Digital Signals

- A digital signal can be transmitted from one point to another in two main ways: **baseband transmission** and **broadband transmission (modulation)**.

- ## Baseband Transmission

- **Baseband transmission** means sending the **digital signal directly** through the transmission medium without converting it into an analog signal. It requires a **low-pass channel**, whose frequency range starts from zero. For example, when two computers are connected using a dedicated cable and the entire bandwidth of the cable is used for a single communication channel, baseband transmission can be used. In simple terms, **digital signal → transmission medium → digital signal**.

- ## Broadband Transmission

- **Broadband transmission**, also called **modulation**, means converting the digital signal into an **analog signal** before transmission. It uses a **band-pass channel**, whose frequency range does not start from zero. During modulation, the digital signal is used to modify a high-frequency **carrier signal**. The resulting signal is a **composite analog signal**suitable for transmission through the channel. At the receiver, the analog signal is converted back into the original digital signal.

Figure 7.11 *Bandwidth of a band-pass channel*

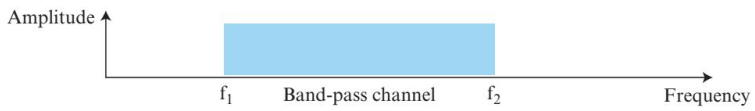
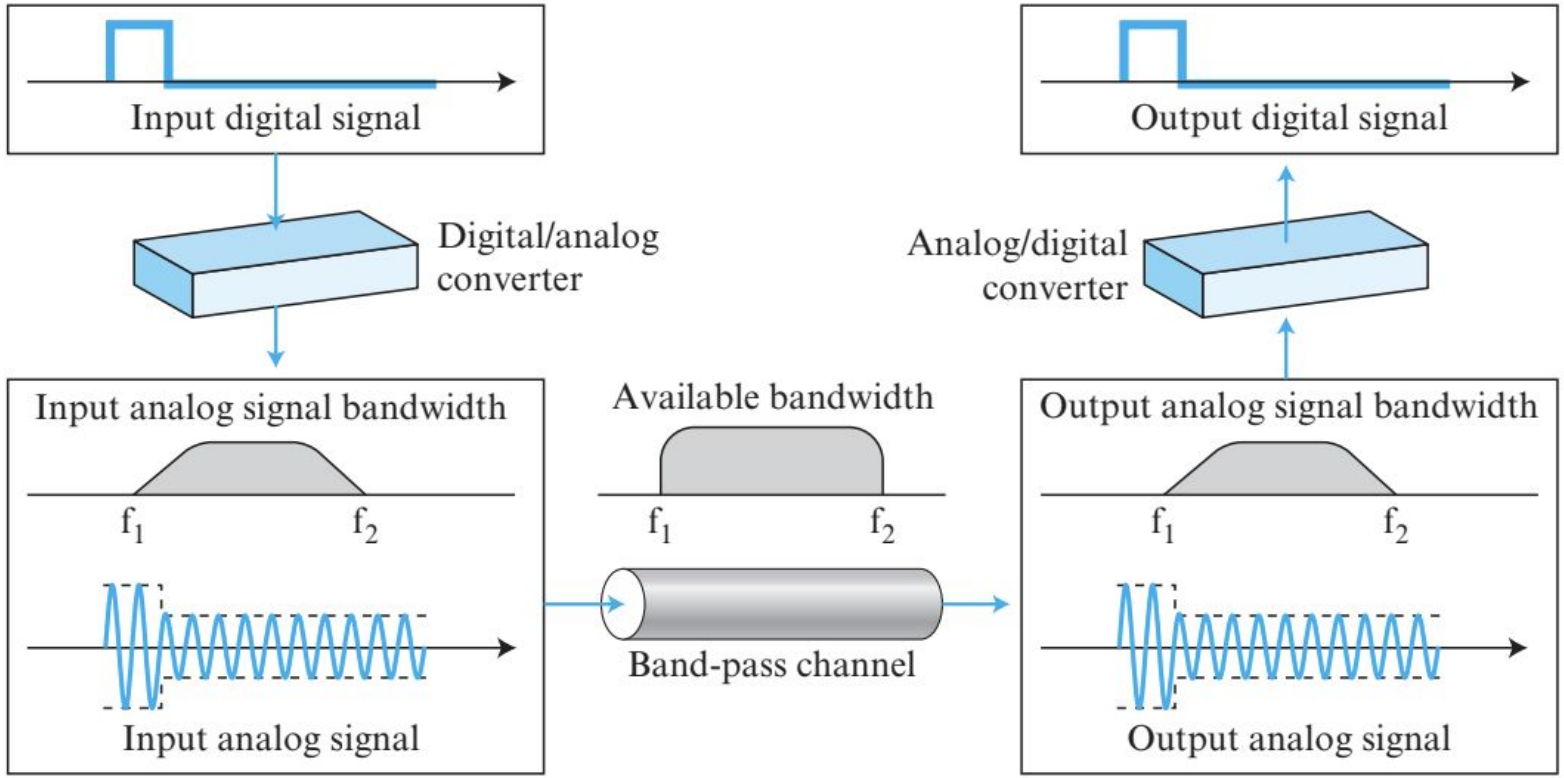


Figure 7.12 *Modulation of a digital signal for transmission on a band-pass channel*



Transmission Impairment

When a signal travels through a transmission medium, the medium is not perfect. As a result, the signal received at the destination may be different from the signal sent by the source. This is called **transmission impairment**.

There are three main causes of impairment:

Attenuation, Distortion, and Noise.

1. Attenuation

Attenuation means **loss of signal energy** as the signal travels through a medium. Some of the signal energy is converted into heat because of the resistance of the medium. **Amplifiers** are used to increase the signal strength and compensate for this loss.

The strength of a signal can be measured using **decibels (dB)**.

$dB = 10 \log_{10} \frac{P_2}{P_1}$

where P1 is the power at the beginning and P2 is the power at the end.

- **Negative dB** → signal is attenuated
- **Positive dB** → signal is amplified

If the power becomes half:

$P_2 = 0.5 P_1$

the loss is approximately **−3 dB**.

2. Distortion

Distortion means a **change in the shape or form of the signal**.

In a composite signal, different frequency components may travel through the medium at different speeds. Therefore, they may reach the destination at different times. This changes their phase and consequently changes the **shape of the original signal**.

3. Noise

Noise is an unwanted signal that can interfere with the original signal.

Common types are:

- **Thermal noise** – caused by random movement of electrons.
- **Induced noise** – caused by electrical devices such as motors and appliances.
- **Crosstalk** – interference from another nearby wire.
- **Impulse noise** – sudden, high-energy spikes caused by sources such as lightning or power lines.

Signal-to-Noise Ratio (SNR)

SNR indicates the relationship between the signal power and noise power.

SNR=Average Signal PowerAverage Noise Power

In decibels:

SNRdB=10log10(SNR)

- **High SNR** → less noise/interference → better signal quality
- **Low SNR** → more noise/interference → poorer signal quality

Nyquist Bit Rate – Noiseless Channel

$$BitRate = 2B \log_2 L$$

where:

- B = bandwidth in Hz
- L = number of signal levels

Nyquist determines the theoretical maximum bit rate for a **noiseless channel**.

Shannon Capacity – Noisy Channel

$$C = B \log_2(1 + SNR)$$

where:

- C = channel capacity in bps
- B = bandwidth
- SNR = signal-to-noise ratio

Shannon gives the **theoretical maximum data rate for a noisy channel**, regardless of the number of signal levels.

Remember:

Nyquist → Noiseless channel

Shannon → Noisy channel

Data Rate Limits

The maximum data rate of a channel depends mainly on:

1. **Bandwidth**
2. **Number of signal levels**
3. **Noise level**

Performance

Important measures of network performance include:

1. Bandwidth

Bandwidth can be expressed in two ways:

- **Bandwidth in Hz** → range of frequencies a channel can carry.
- **Bandwidth in bps** → maximum data transmission rate.

2. Throughput

Throughput is the **actual rate at which data is transmitted**.

For example, a link may have a bandwidth of **1 Mbps**, but if it actually transmits only **200 kbps**, its throughput is 200 kbps.

Bandwidth = Potential capacity

Throughput = Actual data rate

3. Latency (Delay)

Latency is the **total time taken for a message to reach the destination**.

It includes:

- Propagation delay
- Transmission delay
- Queuing delay
- Processing delay

4. Jitter

Jitter occurs when different packets experience **different delays**.

It is especially problematic for **real-time applications** such as audio and video.

Example:
Packet 1 → 20 ms
Packet 2 → 45 ms
Packet 3 → 40 ms

The variation in delay is called **jitter**.

Digital Transmission

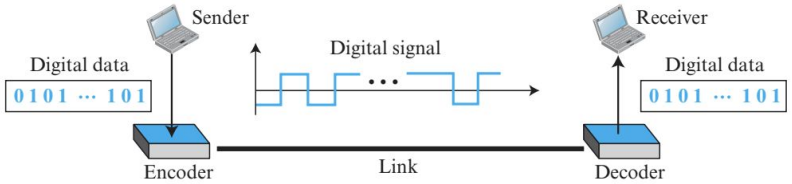
Digital transmission means **converting data into signals for transmission**.

- **Digital data → Digital signal → Digital-to-Digital Conversion**
- **Analog data → Digital signal → Analog-to-Digital Conversion**

For digital-to-digital conversion, three techniques are used:

1. **Line coding** – always required
2. **Block coding** – may be required
3. **Scrambling** – may be required

Figure 7.19 *Line coding and decoding*



Digital-to-Digital Conversion

Line Coding

Line coding is the process of converting a sequence of **digital bits (0s and 1s) into a digital signal**.At the **sender**, bits are encoded into a signal.At the **receiver**, the signal is decoded back into bits.

Line coding schemes are mainly classified into:

- **Polar**
- **Bipolar**
- **Multilevel**

Important Terms

- **Bit rate (N):** Number of bits transmitted per second.
- **Baud rate (S):** Number of signal changes per second.
- **Bits per signal element (r):** Number of bits represented by one signal element.

1. Polar Schemes

In **polar encoding**, the signal uses **positive and negative voltage levels**.

Important polar schemes are:

NRZ-L

The **voltage level represents the bit value**.

NRZ-I

The **change or no change in voltage** represents the bit.

- **Transition → 1**
- **No transition → 0**

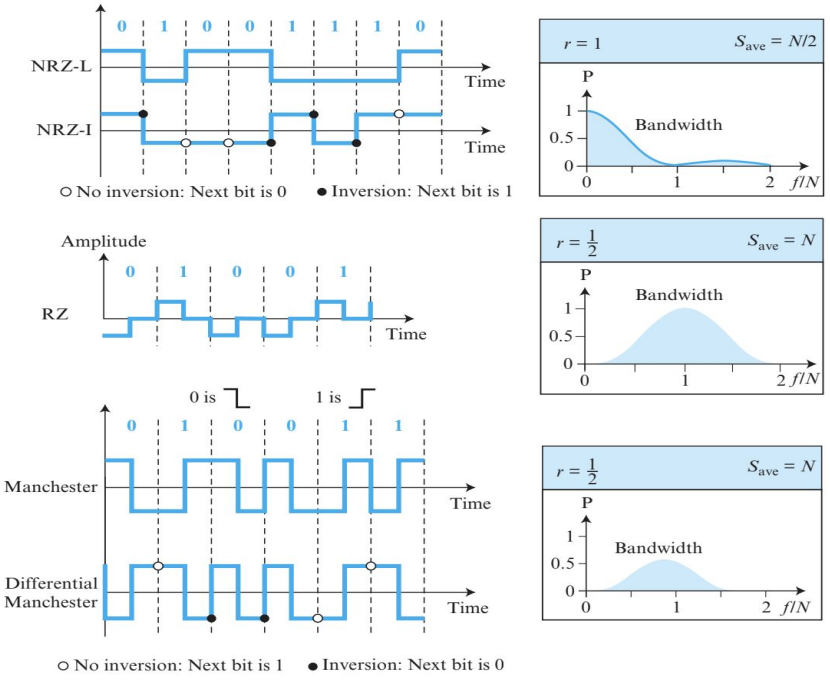
NRZ schemes have a problem with **long sequences of 0s or 1s**, because synchronization becomes difficult.

RZ (Return-to-Zero)

RZ uses **positive, negative, and zero levels**. The signal returns to zero during each bit.

Advantage: Better synchronization
Disadvantage: Requires more bandwidth.

Figure 7.20 *Polar schemes*



Manchester

Manchester combines the ideas of **NRZ-L and RZ**.

There is always a **transition in the middle of every bit**, which provides synchronization.

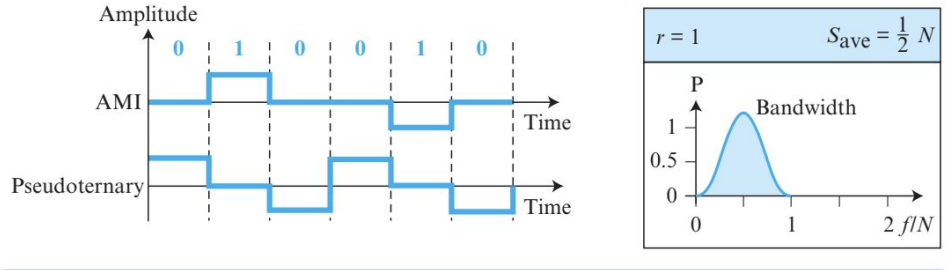
Differential Manchester

There is always a transition in the middle of the bit.

The bit value is determined by the **transition at the beginning**:

- **Transition → 0**
- **No transition → 1**

Figure 7.21 *Bipolar schemes: AMI and pseudoternary*



2. Bipolar Schemes

Bipolar encoding uses **three voltage levels**:

Positive, Negative, and Zero

AMI (Alternate Mark Inversion)

- **0 → Zero voltage**
- **1 → Alternating positive and negative voltage**

Example:

1 → +V
1 → −V
1 → +V

Pseudoternary

It is the opposite of AMI:

- **1 → Zero voltage**
- **0 → Alternating positive and negative voltage**

3. Multilevel Schemes

Multilevel encoding is designed to **increase data speed or reduce bandwidth**.

It encodes **multiple bits into fewer signal elements**.

The general notation is:**mBnL**

- **m** → number of input bits
- **B** → binary data
- **n** → number of signal elements
- **L** → number of signal levels

Examples:**2B1Q**

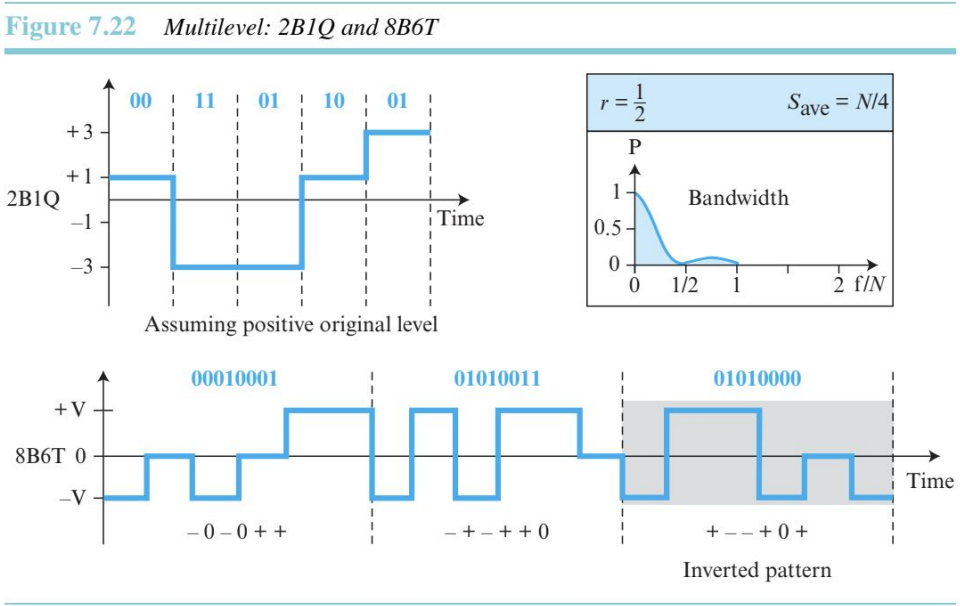
2 bits → 1 signal element

Uses **4 voltage levels** (quaternary).

8B6T 8 bits → 6 signal elements

Uses **3 voltage levels** (ternary).

8B6T provides additional signal patterns that can help with **synchronization and error detection** and is used with **100BASE-4T**.



Block Coding and Scrambling

1. Block Coding

Block coding adds some extra bits (redundancy) to the original data. These extra bits help to:

- Maintain **synchronization** between sender and receiver.
- Provide **error detection**.
- Improve the performance of line coding.

In block coding, a block of **m bits is changed into n bits**, where **n > m**. It is represented as **mB/nB**. For example, **4B/5B**means **4 input bits are converted into 5 output bits**.

Steps in Block Coding

Block coding has three main steps:

1. **Division** – Divide the original bit stream into groups of **m bits**.
2. **Substitution** – Replace each m-bit group with an **n-bit code**.
3. **Combination** – Combine all the n-bit groups to form the new bit stream.

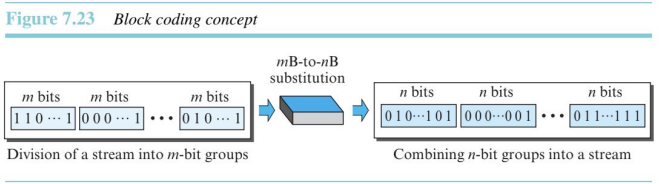


Figure 7.23 *Block coding concept*

Example:
In **4B/5B**, every 4-bit group is replaced by a 5-bit group.

So:**4 bits → 5 bits → more redundancy → better synchronization and error detection**

2. 4B/5B Coding

4B/5B was designed mainly to solve the synchronization problem of **NRZ-I**.

In NRZ-I, a long sequence of **0s** can cause the receiver to lose synchronization because there are no signal transitions for a long time.

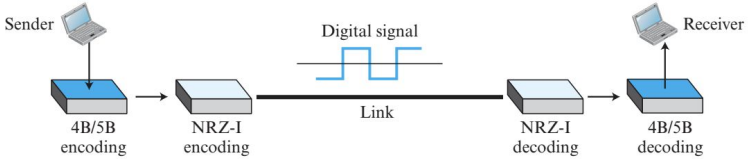
4B/5B solves this by converting every **4-bit group into a 5-bit code** in such a way that the output does not contain a long sequence of zeros.

Important point

The 5-bit code is designed to have:

- Maximum **one leading 0**
- Maximum **two trailing 0s**

Figure 7.24 *Using block coding 4B/5B with NRZ-I line coding scheme*



Therefore, when different 5-bit groups are combined, there can be **at most three consecutive 0s**.

At the receiver:

NRZ-I signal → decode into 5-bit groups → remove extra bits → original 4-bit data

Main purpose of 4B/5B

Avoid long sequences of 0s and maintain synchronization.

3. 8B/10B Coding

8B/10B is similar to 4B/5B, but: **8 data bits → 10-bit code**

It provides **better error detection and synchronization** than 4B/5B.

The 8 bits are divided into:

- **5 bits → 5B/6B**
- **3 bits → 3B/4B**

Therefore:**5B/6B + 3B/4B = 8B/10B**

Disparity Controller

8B/10B uses a **disparity controller** to avoid long sequences of 0s or 1s.

It keeps track of whether the previous blocks contain more:

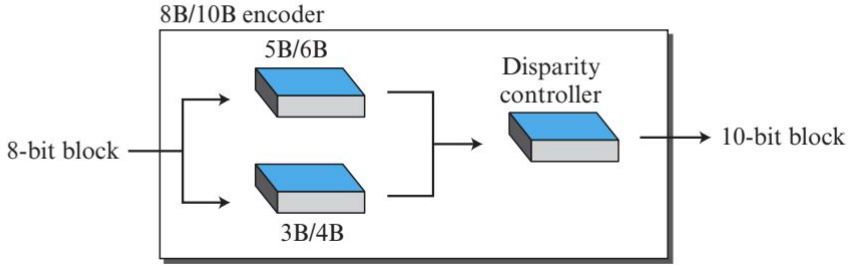
- 0s than 1s, or
- 1s than 0s.

If necessary, the bits in the current code are complemented:

0 → 1
1 → 0

This helps maintain a balance between 0s and 1s.

Figure 7.25 *8B/10B block encoding*



here are:

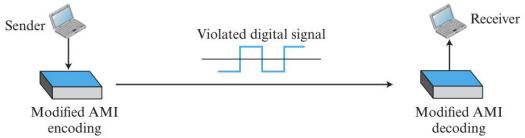
$2^{10} - 2^8 = 768$ redundant codes

These redundant codes can be used for **error detection and disparity checking**.

Main purpose of 8B/10B

Better synchronization + better error detection than 4B/5B.

Figure 7.26 *AMI used with scrambling*



4. Scrambling

Scrambling is used mainly with **bipolar AMI** for long-distance communication.

Why is scrambling needed?

AMI has an advantage:

- It has **narrow bandwidth**.
- It does **not produce a DC component**.

But AMI has one major problem:

A long sequence of 0s produces no signal transition → synchronization is lost.

We need a technique that:

- Avoids long sequences of 0s.
- Maintains synchronization.
- Does **not increase the number of bits**.
- Maintains the advantages of AMI.

This technique is called **scrambling**.

Block Coding vs Scrambling

Block Coding	Scrambling
Adds extra bits	Does not add extra bits
Increases number of bits	Bit rate remains the same
Done before line coding	Done during encoding
Example: 4B/5B, 8B/10B	Example: B8ZS, HDB3

Two common scrambling techniques are:

B8ZS and **HDB3**

SCMS SCHOOL OF ENGINEERING & TECHNOLOGY
Vidya Nagar, Palissery, Karukutty, Ernakulam - 683576,Kerala, India.
www.scmsgroup.org/sset

5. B8ZS Coding

B8ZS = Bipolar with 8-Zero Substitution

It is commonly used in **North America**.

In B8ZS, whenever **eight consecutive 0s** occur, they are replaced by:

000VB0VB

Where:

- **B = Bipolar pulse** that follows the normal AMI rule.
- **V = Violation pulse** that intentionally violates the AMI rule.

The purpose is to create signal transitions and therefore maintain synchronization.

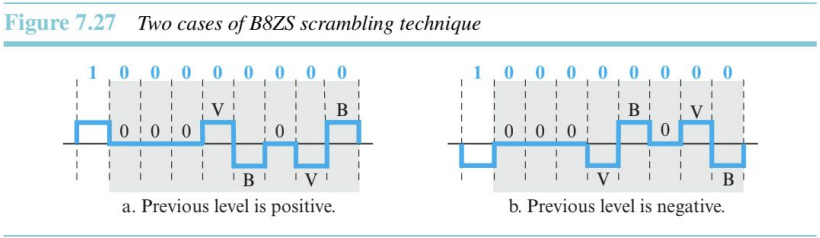


Figure 7.27 *Two cases of B8ZS scrambling technique*

Important points

- 8 zeros are replaced by 8 pulses/bits.
- **Bit rate does not change.**
- Synchronization is maintained.
- Positive and negative pulses remain balanced, so **DC balance is maintained**.

Main idea:

8 zeros → 000VB0VB → signal transitions → synchronization

6. HDB3 Coding

HDB3 = High-Density Bipolar 3-Zero

It is commonly used **outside North America**.

In HDB3, whenever **four consecutive 0s** occur, they are replaced by either:

000V or **B00V**

The choice depends on the number of nonzero pulses since the previous substitution.

Rule

- If the number of nonzero pulses is **odd** → use **000V**.
- If the number of nonzero pulses is **even** → use **B00V**.

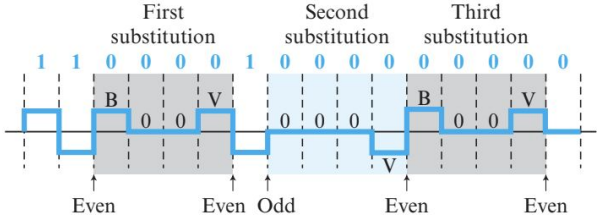
This helps maintain the required balance of positive and negative pulses.

Example idea

Four 0s → 000V or B00V

The selection depends on the previous number of nonzero pulses.

Figure 7.28 *Different situations in HDB3 scrambling technique*



Technique	Main Idea	Important Point
Block Coding	m bits → n bits	Adds redundancy
4B/5B	4 bits → 5 bits	Prevents long sequences of 0s
8B/10B	8 bits → 10 bits	Better synchronization and error detection
Scrambling	Modify AMI encoding	Does not add bits
B8ZS	8 zeros → 000VB0VB	Common in North America
HDB3	4 zeros → 000V or B00V	Common outside North America

Analog-to-Digital Conversion : Sometimes the input to a communication system is an **analog signal**, such as voice from a microphone or an image from a camera. Since digital signals are generally less affected by noise, analog signals are often converted into digital data. This process is called **analog-to-digital conversion or digitization**. Two important techniques are **Pulse Code Modulation (PCM)** and **Delta Modulation (DM)**. After converting the analog signal into digital data, digital-to-digital techniques such as line coding can be used to convert the data into a digital signal.

Pulse Code Modulation (PCM) :Pulse Code Modulation (PCM) is the most common technique used to convert an analog signal into digital data. PCM has three main steps: **sampling, quantization, and encoding**. In sampling, the analog signal is measured at regular time intervals. In quantization, the sampled amplitude values are approximated to a fixed number of levels. Finally, in encoding, each quantized value is converted into a binary code.

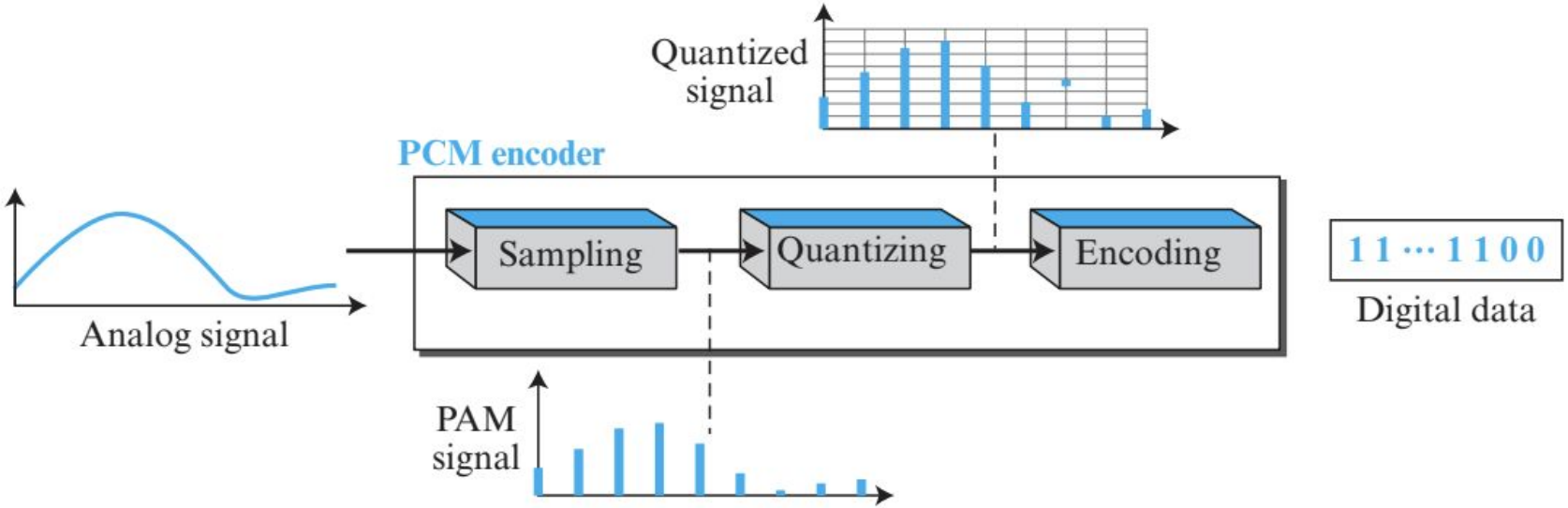
Sampling :Sampling means measuring the amplitude of an analog signal at regular intervals. If the samples are taken every **Ts seconds**, Ts is called the **sampling interval**. The number of samples taken per second is called the **sampling rate or sampling frequency**, represented by **fs**.

[
f_s = \frac{1}{T_s}
]

The sampling process can be viewed as **Pulse Amplitude Modulation (PAM)** because the amplitude of each pulse represents the sampled value. The most commonly used method is **sample-and-hold**, where the sampled amplitude is held constant for a short period, producing flat-top samples.

Figure 7.29:

Figure 7.29 *Components of PCM encoder*



Sampling Theorem

The **Nyquist sampling theorem** states that, to reproduce the original analog signal correctly, the sampling rate must be at least **twice the highest frequency** present in the signal.

$$
f_s \geq 2f_{max}
$$

The minimum sampling rate is called the **Nyquist rate**.

For example, human voice generally contains frequencies up to **4000 Hz**. Therefore,

$$
f_s = 2 \times 4000 = 8000\ \text{samples/s}
$$

So, the voice signal must be sampled at least **8000 times per second**.

Figure 7.30 *Three different sampling methods for PCM*

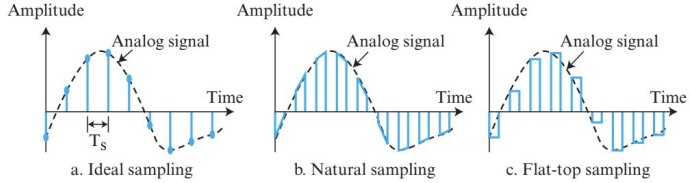
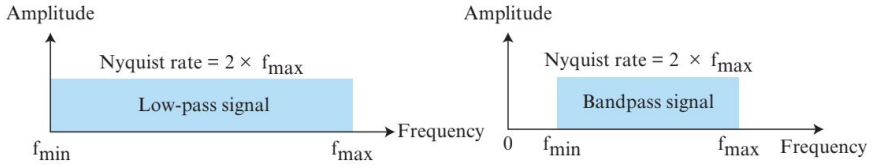


Figure 7.31 *Nyquist sampling rate for low-pass and bandpass signals*



Quantization

After sampling, the amplitude of each sample can have many possible values. These values must be converted into a fixed number of levels before they can be represented digitally. This process is called **quantization**. First, the amplitude range from **Vmin to Vmax** is divided into **L quantization levels**. Each level represents a particular range of amplitudes, and the actual sample value is approximated to the nearest quantized value.

The size of each quantization interval is:

$$
\Delta = \frac{V_{max}-V_{min}}{L}
$$

For example, if the signal varies from **−20 V to +20 V** and we use **8 levels**, then:

$$
\Delta = \frac{20-(-20)}{8}=\frac{40}{8}=5V
$$

Thus, the amplitude range is divided into eight levels, and every sample is assigned to one of these levels.

Quantization Error

Quantization is an approximation process, so the original sample value and the quantized value may not be exactly the same. The difference between them is called **quantization error**. The maximum quantization error is less than half of the quantization interval.

$$
-\frac{\Delta}{2} \leq Error \leq \frac{\Delta}{2}
$$

Increasing the number of quantization levels reduces the quantization error and improves the quality of the reconstructed signal. However, more levels require more bits per sample.

The approximate signal-to-noise ratio due to quantization is:

$$
SNR_{dB}=6.02n_b+1.76\ dB
$$

where **nb** is the number of bits used for each sample.

Quantization Levels and Bits per Sample

The number of bits required to represent each sample depends on the number of quantization levels **L**.

$$
n_b=\log_2 L
$$

For example, if **L = 8**:

$$
n_b=\log_2 8=3\ bits
$$

Therefore, each sample is represented using **3 bits**. In audio digitization, **256 levels** are commonly used, while video may require thousands of levels because it contains much more variation.

Encoding

The final step in PCM is **encoding**. After each sample has been quantized, its quantization level is represented using a binary code. If there are **L quantization levels**, the number of bits required for each sample is:

$$
n_b=\log_2 L
$$

For example, with 8 levels, 3 bits are required. A quantization level of 2 can be represented as **010**, while level 5 can be represented as **101**.

The PCM bit rate is calculated as:

$$
Bit\ Rate = Sampling\ Rate \times Bits\ per\ Sample
$$

or

$$
Bit\ Rate=f_s\times n_b
$$

Example: Human Voice

Human voice normally contains frequencies up to **4000 Hz**. According to the Nyquist theorem:

$$
f_s=2\times4000=8000\ samples/s
$$

If each sample requires **8 bits**:

$$
Bit\ Rate=8000\times8=64,000\ bps
$$

Therefore, the required bit rate is:

$$
\boxed{64\ kbps}
$$

Recovery of the Original Signal

At the receiver, a **PCM decoder** converts the received binary code words back into corresponding amplitude values. These values form a staircase-like signal. A **low-pass filter** is then used to smooth this staircase signal and reconstruct the original analog signal. If the signal is sampled at or above the Nyquist rate and a sufficient number of quantization levels are used, the original signal can be reconstructed with good accuracy.

PCM Bandwidth

Digitization requires more bandwidth because the analog signal is converted into a high-rate digital signal. The minimum bandwidth of the digital signal is approximately:

[
B_{min}=n_b\times B_{analog}
]

where **nb** is the number of bits per sample and **Banalog** is the bandwidth of the original analog signal.

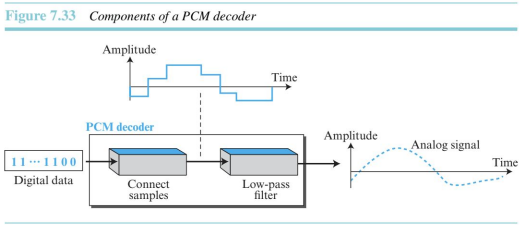


Figure 7.33 *Components of a PCM decoder*

Thus, using more bits per sample improves accuracy but also increases the required digital bit rate and bandwidth.

Delta Modulation (DM)

Delta Modulation (DM) is a simpler alternative to PCM. PCM records the **actual amplitude value** of every sample, whereas Delta Modulation records only the **change in amplitude between the current sample and the previous sample**. Therefore, DM does not send a complete code word for every sample; instead, it sends **one bit at a time** to indicate whether the signal has increased or decreased.

In simple terms, **PCM asks: "What is the amplitude?"**, while **DM asks: "Did the amplitude increase or decrease?"**Because of this simpler approach, DM requires less complex hardware than PCM, although it can introduce errors when the signal changes very rapidly.

Easy Flow

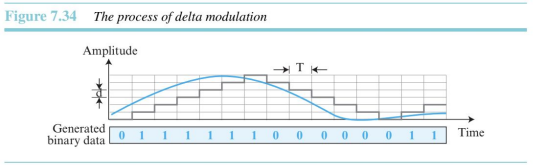
Analog Signal → Sampling → Quantization → Encoding → Digital Data

For PCM:

Analog signal → Sample → Quantize → Encode → Bits

For recovery:

Bits → Decode → Staircase signal → Low-pass filter → Original analog signal



7.3 ANALOG TRANSMISSION

Analog transmission is used when the available communication channel is a **bandpass channel**. Digital transmission generally requires a low-pass channel, but when a bandpass channel is available, digital data can be converted into an analog signal before transmission. This process is called **digital-to-analog conversion**. Similarly, when an analog low-pass signal needs to be transmitted through a bandpass channel, it is converted into another analog signal. This process is called **analog-to-analog conversion**.

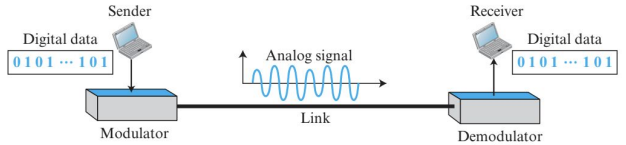
7.3.1 Digital-to-Analog Conversion

Digital-to-analog conversion is the process of converting **digital data (0s and 1s)** into an **analog signal** suitable for transmission through a bandpass channel.

An analog sine wave has three basic characteristics:

1. **Amplitude** – the height or strength of the signal.
2. **Frequency** – the number of cycles completed per second.
3. **Phase** – the position of the waveform with respect to time.

Figure 7.35 *Digital-to-analog conversion*



If we change any one of these characteristics according to the digital data, we can represent 0s and 1s using an analog signal.

The three basic digital-to-analog modulation techniques are:

- **Amplitude Shift Keying (ASK)** – changes the amplitude.
- **Frequency Shift Keying (FSK)** – changes the frequency.
- **Phase Shift Keying (PSK)** – changes the phase.

There is also a more efficient technique called **Quadrature Amplitude Modulation (QAM)**, which changes both **amplitude and phase**.

SCMS SCHOOL OF ENGINEERING & TECHNOLOGY
Vidya Nagar, Palissery, Karukutty, Ernakulam - 683576,Kerala, India.
www.scmsgroup.org/sset

Think of a sine wave as a signal that can be controlled in three ways:

ASK → Change height
FSK → Change speed
PSK → Change position
QAM → Change height + position

1. Amplitude Shift Keying (ASK): In **Amplitude Shift Keying (ASK)**, the **amplitude of the carrier signal is changed** according to the digital data.

The **frequency and phase remain constant**.

For example:

- Bit **1** → **higher/full amplitude**
- Bit **0** → **zero or lower amplitude**

The simplest form is called **Binary ASK (BASK)** or **On-Off Keying (OOK)**.

In OOK, one signal level has an amplitude of **0**, while the other level has the amplitude of the carrier signal.

Example

If the data is:

1 0 1 1 0

the carrier signal is switched:

ON → OFF → ON → ON → OFF

Therefore, ASK can be understood as:

> **ASK = Change the amplitude to represent bits.**

ASK Bandwidth

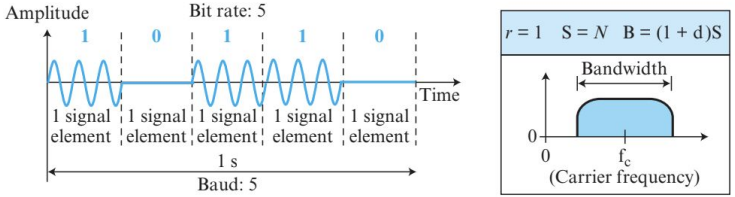
The bandwidth of ASK depends on the signal rate and the filtering/modulation factor.

B=(1+d)S

where:

- **B** = bandwidth
- **S** = signal rate (baud rate)
- **d** = factor related to modulation and filtering
- 0≤d≤1

Figure 7.36 *Binary amplitude shift keying*



Therefore, the minimum bandwidth is approximately **S**, and the maximum can approach **2S**.

The important advantage is that the carrier frequency fc can be selected according to the available bandpass channel.

Multilevel ASK

ASK does not have to use only two amplitude levels. We can use **multiple amplitude levels**.

For example, we can use:

- 4 amplitude levels → 2 bits at a time
- 8 amplitude levels → 3 bits at a time
- 16 amplitude levels → 4 bits at a time

Therefore, more bits can be transmitted using one signal element.

However, practical high-level amplitude modulation is generally implemented using **QAM** rather than pure ASK.

2. Frequency Shift Keying (FSK)

In **Frequency Shift Keying (FSK)**, the **frequency of the carrier signal is changed** according to the digital data.

The **amplitude and phase remain constant**.

In **Binary FSK (BFSK)**, two different frequencies are used:

- Frequency f1 → represents **0**
- Frequency f2 → represents **1**

For example, if the data is:

1 0 1 1 0

the transmitter changes between the two frequencies according to each bit.

Therefore:

> **FSK = Change the frequency to represent bits.**

Easy way to remember

Imagine a person saying:

0 → slow voice
1 → fast voice

The amplitude remains approximately the same, but the frequency changes.

FSK Bandwidth :

The bandwidth required for FSK is:

$B=(1+d)S+2\Delta f$

where:

- **B** = bandwidth
- **S** = signal rate
- **d** = filtering/modulation factor
- $2\Delta f$ = difference between the two carrier frequencies.

For proper operation, the frequency separation must be sufficiently large.

Multilevel FSK :

Instead of using only two frequencies, we can use many frequencies.

For example:

- **4 frequencies → 2 bits at a time**
- **8 frequencies → 3 bits at a time**
- **16 frequencies → 4 bits at a time**

This is called **Multilevel FSK (MFSK)**.

The important point is that the different frequencies must be sufficiently separated so that the receiver can distinguish them correctly.

3. Phase Shift Keying (PSK)

In **Phase Shift Keying (PSK)**, the **phase of the carrier signal is changed** according to the digital data.

The **amplitude and frequency remain constant**.

Therefore:

PSK = Change the phase to represent bits.

Binary PSK (BPSK)

The simplest form of PSK is **Binary PSK (BPSK)**.

BPSK uses two phases:

- **0° → one bit**
- **180° → the other bit**

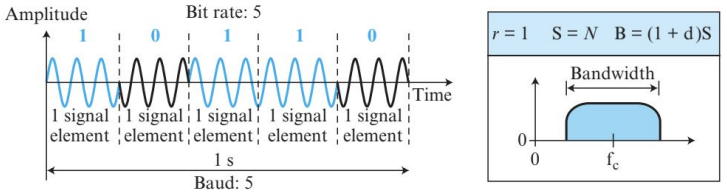
For example:

0 → 0° phase

1 → 180° phase

The phase changes whenever the bit changes.

Figure 7.38 *Binary phase shift keying*



Why is PSK better than ASK in terms of noise?

In ASK, the receiver determines the bit mainly from the **amplitude**. Noise can easily affect the amplitude.

In PSK, the receiver determines the bit mainly from the **phase**. Amplitude changes caused by noise are less directly important for detecting the phase.

Therefore, **BPSK is generally less susceptible to amplitude noise than ASK**.

Quadrature PSK (QPSK)

QPSK is an improved form of PSK that sends **2 bits in one signal element**.

Instead of representing only two possibilities, QPSK provides **four possible phase states**.

Since:

$$2^2 = 4$$

two bits can be represented using four different phase values.

For example:

Bits	Phase
00	0°
01	90°
10	180°
11	270°

The major advantage of QPSK is that **2 bits are transmitted per signal element**, so the baud rate can be reduced for a given bit rate.

Constellation Diagram

A **constellation diagram** is a graphical method used to represent the different signal elements used in modulation schemes such as **PSK, ASK, and QAM**.

It has two axes:

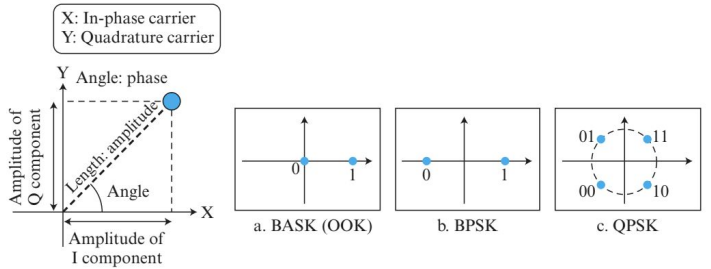
- **X-axis → In-phase (I) component**
- **Y-axis → Quadrature (Q) component**

Each possible signal is represented by a **point**.

From a point in the constellation diagram, we can determine:

- **Amplitude** → distance from the origin
- **Phase** → angle from the X-axis
- **I component** → position along X-axis
- **Q component** → position along Y-axis

Figure 7.39 *Concept of a constellation diagram*



Easy way to remember

Distance from centre = Amplitude

Angle from X-axis = Phase

4. Quadrature Amplitude Modulation (QAM)

QAM combines the advantages of **ASK and PSK**.

In QAM, both:

- **Amplitude**
- **Phase**

are changed.

Therefore:

QAM = Change amplitude + change phase.

QAM uses two carriers:

- **In-phase carrier (I)**
- **Quadrature carrier (Q)**

These carriers are separated by **90°**.

QAM can provide many different signal combinations.

For example:

- **4-QAM**
- **16-QAM**
- **64-QAM**
- **256-QAM**

As the number of signal points increases, more bits can be transmitted per signal element.

For example:

4−QAM=2 bits/signal element16−QAM=4 bits/signal element64−QAM=6 bits/signal element

QAM is widely used because it can achieve **high data rates efficiently**.

Technique	What changes?	What remains constant?
ASK	Amplitude	Frequency and phase
FSK	Frequency	Amplitude and phase
PSK	Phase	Amplitude and frequency
QAM	Amplitude + Phase	Frequency

7.3.2 Analog-to-Analog Conversion

Analog-to-analog conversion, also called **analog modulation**, is the process of representing an analog information signal using another analog signal.

We may ask:

> **If the original information is already analog, why do we need modulation?**

The reason is that many communication channels are **bandpass channels**.

For example, consider **radio broadcasting**. Many radio stations generate low-pass audio signals. If all stations transmitted their signals in the same frequency range, they would interfere with each other.

Therefore, each radio station is assigned a different **carrier frequency**. The audio signal is shifted to that frequency using modulation.

There are three basic analog-to-analog modulation techniques:

1. **Amplitude Modulation (AM)**
2. **Frequency Modulation (FM)**
3. **Phase Modulation (PM)**

1. Amplitude Modulation (AM) : In **Amplitude Modulation (AM)**, the **amplitude of the carrier signal changes according to the amplitude of the information signal**.

The:

- frequency remains constant
- phase remains constant
- amplitude changes

The information signal forms the **envelope** of the carrier.

If the information signal becomes stronger → carrier amplitude becomes larger.

If the information signal becomes weaker → carrier amplitude becomes smaller.

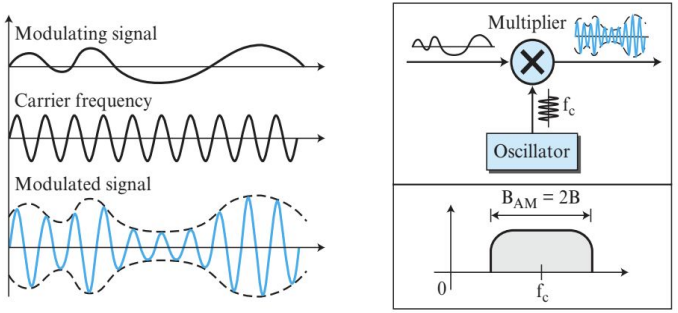
Therefore:

AM = Change the amplitude according to the analog information signal.

AM Bandwidth : If the bandwidth of the original information signal is **B**, the AM signal normally requires:

BAM=2B The resulting bandwidth is centred around the carrier frequency fc.

Figure 7.41 *Amplitude modulation*



2. Frequency Modulation (FM)

In **Frequency Modulation (FM)**, the **frequency of the carrier signal changes according to the amplitude of the information signal**.

The:

- amplitude remains constant
- phase changes as a consequence of frequency variation
- frequency changes

When the amplitude of the information signal increases, the carrier frequency changes accordingly.

Therefore:**FM = Change the frequency according to the analog information signal.**

FM can be generated using a **Voltage-Controlled Oscillator (VCO)**.

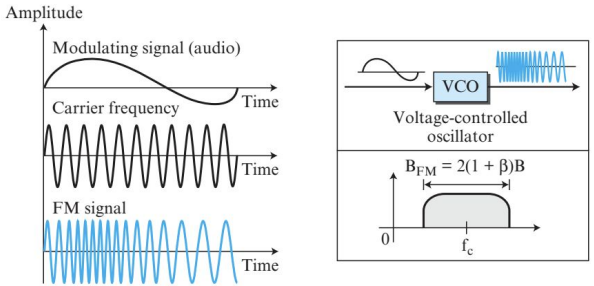
FM Bandwidth : An empirical expression is:

$BFM=2(1+\beta)B$

where:

- **B** = bandwidth of the original analog signal
- **β** = factor related to the modulation technique.

Figure 7.42 *Frequency modulation*



3. Phase Modulation (PM)

In **Phase Modulation (PM)**, the **phase of the carrier signal changes according to the amplitude of the information signal**.

The:

- amplitude remains constant
- frequency remains constant
- phase changes

Therefore:

PM = Change the phase according to the analog information signal.

FM and PM are closely related.

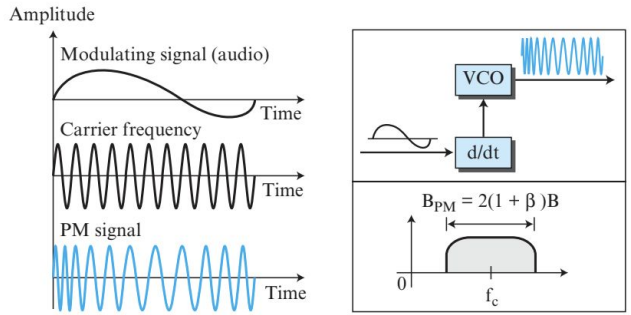
The important difference is:

FM: instantaneous frequency change depends on the amplitude of the information signal.

PM: instantaneous frequency change depends on the **derivative** of the information signal.

PM generally requires a bandwidth several times that of the original analog signal.

Figure 7.43 *Phase modulation*



Technique	What changes?	Amplitude	Frequency
AM	Amplitude	Changes	Constant
FM	Frequency	Constant	Changes
PM	Phase	Constant	Constant*

BANDWIDTH UTILIZATION

Bandwidth is a limited resource in communication. **Bandwidth utilization** means using the available bandwidth efficiently. Two major techniques are:

1. **Multiplexing** – combines multiple signals into one link to improve **efficiency**.
2. **Spread Spectrum** – increases the bandwidth of a signal to provide **privacy and protection against interference/jamming**.

1. MULTIPLEXING

Multiplexing allows multiple signals to share a single high-bandwidth communication link.

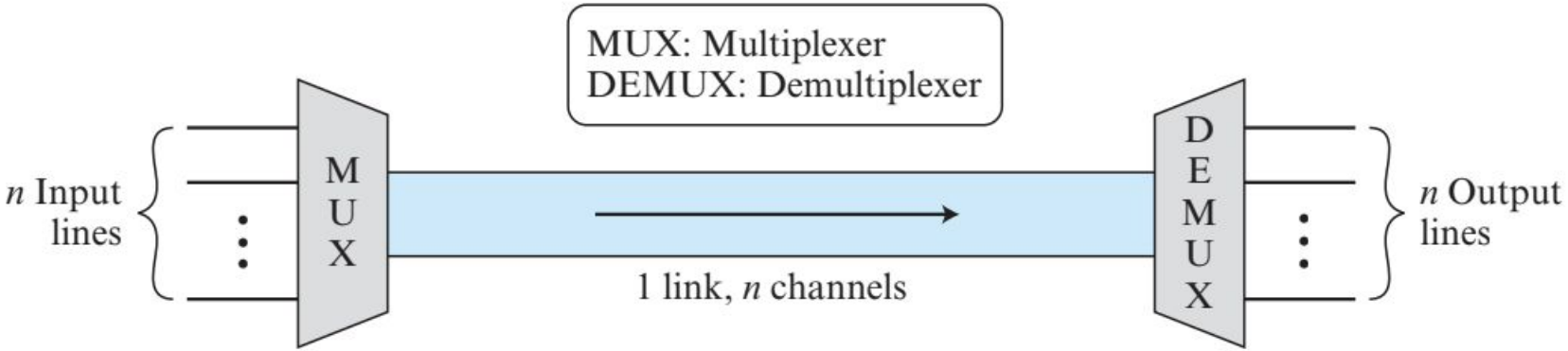
Basic idea:

Many signals → Multiplexer → One link → Demultiplexer → Separate signals

A **multiplexer (MUX)** combines signals, while a **demultiplexer (DEMUX)** separates them at the receiver.

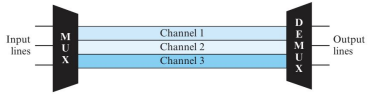
Type	Main idea	Signal
FDM	Share bandwidth by frequency	Analog
WDM	Share fiber using different light wavelengths	Optical
TDM	Share the link by time	Digital

Figure 7.44 *Dividing a link into channels*



FDM – Frequency-Division Multiplexing

Figure 7.45 *Frequency-division multiplexing*



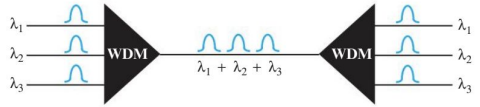
In **FDM**, the available bandwidth is divided into different **frequency bands**. Each signal uses a different carrier frequency.

Example:
Signal 1 → 100 kHz
Signal 2 → 200 kHz
Signal 3 → 300 kHz Small unused frequency ranges called **guard bands** prevent signals from interfering with each other.

WDM – Wavelength-Division Multiplexing;

WDM is mainly used with **fiber-optic cables**. Different signals are transmitted using different **wavelengths of light**.

Figure 7.47 *Wavelength-division multiplexing*



Many light signals → MUX → Single optical fiber → DEMUX → Separate signals

WDM is conceptually similar to FDM, but it uses **light wavelengths**.

TDM – Time-Division Multiplexing:

In **TDM**, users share the same link by taking turns in **different time slots**.

User 1 → User 2 → User 3 → User 4 → User 1 → ...

Figure 7.48 *TDM*



There are two types:

- **Synchronous TDM:** Each input gets a fixed time slot, even if it has no data.
- **Statistical TDM:** Time slots are given only to inputs that have data, so bandwidth is used more efficiently.

Simple difference:

Synchronous TDM = fixed slots
Statistical TDM = slots given when needed

2. SPREAD SPECTRUM

Spread Spectrum (SS) increases the bandwidth of a signal much more than its original requirement.

Purpose:

- Improve **privacy**
- Reduce the effect of **interference**
- Provide **anti-jamming** capability

If the original bandwidth is **B**, spread spectrum uses a much larger bandwidth:

BSS >> B

There are two main techniques:

1. FHSS – Frequency Hopping Spread Spectrum

In **FHSS**, the signal rapidly **jumps from one frequency to another** according to a pseudorandom sequence.

Frequency 1 → Frequency 3 → Frequency 2 → Frequency 4 → ...

The sender and receiver know the same hopping sequence.

Advantages:

- Difficult for an unauthorized user to follow.
- Reduces the effect of interference and jamming.

Figure 7.53 *Spread spectrum*

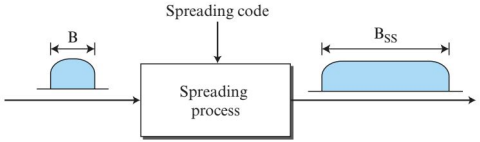
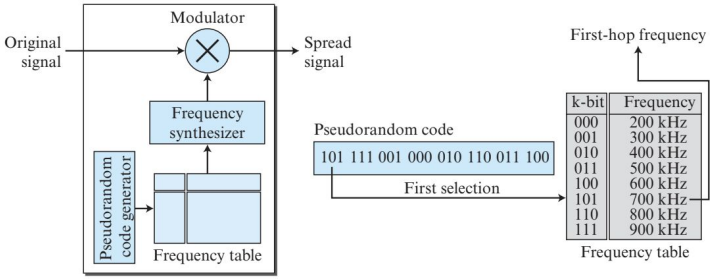


Figure 7.54 *Frequency hopping spread spectrum (FHSS)*



2. DSSS – Direct Sequence Spread Spectrum

In **DSSS**, each data bit is replaced by a group of smaller bits called **chips** using a spreading code.

For example:

1 data bit → 11 chips

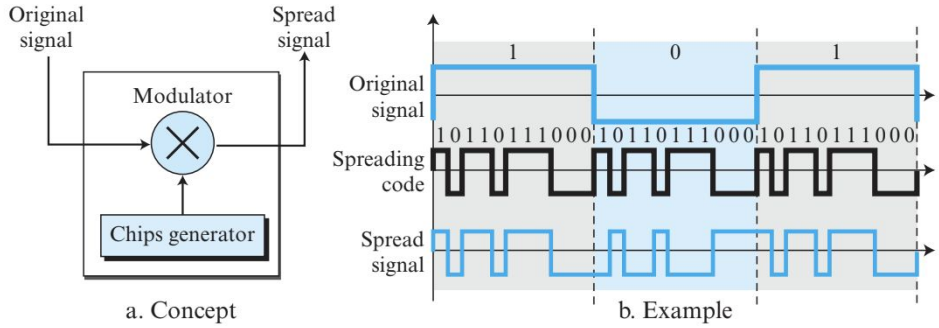
If the original data rate is **N**, using 11 chips makes the spread signal rate:

11N

Therefore, the required bandwidth becomes approximately **11 times larger**.

Multiplexing	Spread Spectrum
Combines multiple signals	Expands signal bandwidth
Main goal: **efficiency**	Main goals: **privacy & anti-jamming**
Shares available bandwidth	Uses much larger bandwidth
FDM, WDM, TDM	FHSS, DSSS

Figure 7.57 *DSSS*



7.5 TRANSMISSION MEDIA

A **transmission medium** is the path through which information travels from a sender to a receiver. In data communication, the medium may be a **metallic cable, fiber-optic cable, or free space**. Transmission media are located below the physical layer and are directly controlled by it; therefore, they can be considered as **Layer 0**.

Transmission media are broadly divided into **guided media** and **unguided media**. Guided media use a physical path such as a cable, while unguided media transmit signals through free space.

7.5.1 Guided Media

Guided media provide a physical path for signals. The main types are **twisted-pair cable, coaxial cable, and fiber-optic cable**.

Twisted-Pair Cable

A **twisted-pair cable** consists of two insulated copper wires twisted together. Twisting helps reduce **noise and crosstalk**. The two common types are **UTP (Unshielded Twisted Pair)** and **STP (Shielded Twisted Pair)**. STP has an additional metallic shield that provides better protection from interference but makes the cable more expensive and bulky.

Twisted-pair cables are commonly used in **telephone lines, DSL connections, and Ethernet LANs**. Their attenuation increases as frequency increases.

Figure 7.59 *Twisted-pair cable*

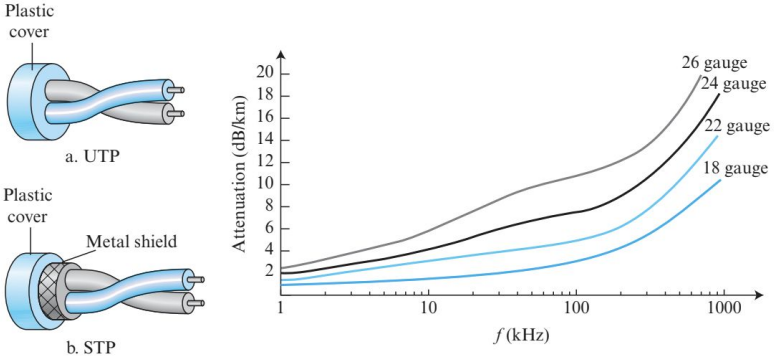


Figure 7.58 *Transmission media and physical layer*

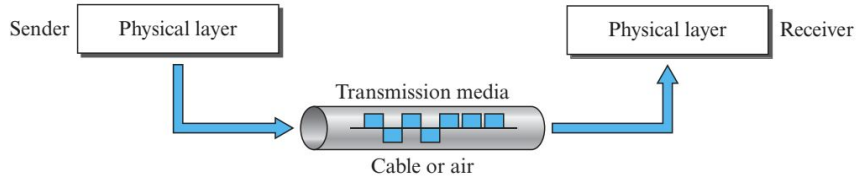
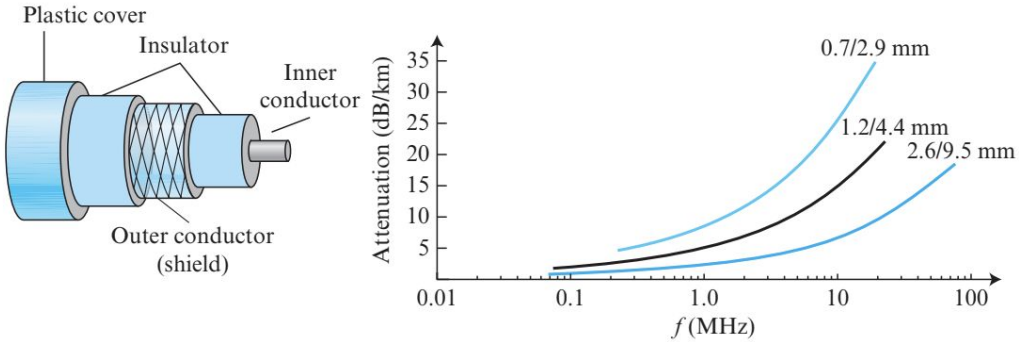
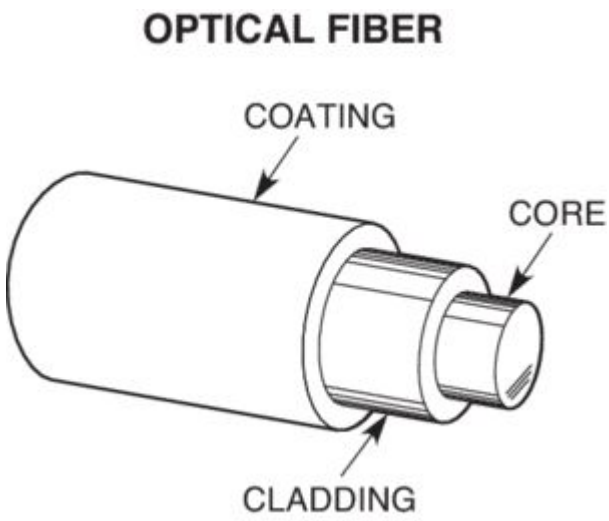


Figure 7.60 *Coaxial cable*





SCMS SCHOOL OF ENGINEERING & TECHNOLOGY
Vidya Nagar, Palissery, Karukutty, Ernakulam - 683576,Kerala, India.
www.scmsgroup.org/sset

Coaxial Cable

A **coaxial cable** consists of a central copper conductor surrounded by insulation, a metallic shield, another insulating layer, and an outer protective cover. The metallic shield protects the signal from interference and also acts as the second conductor.

Coaxial cable provides **higher bandwidth than twisted pair** and has good resistance to noise. It has been used in **telephone networks, cable TV networks, and early Ethernet LANs**. Modern telephone networks have largely replaced coaxial cable with fiber optics.

Fiber-Optic Cable

A **fiber-optic cable** carries information as **light signals** through very thin glass or plastic fibers. It provides **very high bandwidth, low attenuation, long-distance communication, and excellent resistance to electromagnetic interference**.

Fiber-optic cable is widely used in **Internet backbones, telephone networks, cable TV, data centers, and submarine communication**.

7.5.2 Unguided Media

In unguided media, signals travel through **free space without a physical cable**. The major types are **radio waves, microwaves, and infrared waves**.

Radio Waves

Radio waves are generally **omnidirectional**, meaning they travel in many directions. Therefore, the transmitting and receiving antennas do not need to be precisely aligned. Radio waves can travel long distances and some can penetrate walls. They are commonly used in **radio broadcasting and wireless communication**. However, they are susceptible to interference, and many frequency bands are regulated by authorities.

Microwaves

Microwaves have frequencies approximately between **1 and 300 GHz**. They are generally **unidirectional** and require **line-of-sight communication**. The transmitting and receiving antennas must be aligned. Tall towers and repeaters may be required for long-distance communication.

Microwaves are widely used in **cellular communication, satellite communication, wireless LANs, and point-to-point links**. Their large bandwidth allows high data rates.

Infrared

Infrared waves have frequencies from approximately **300 GHz to 400 THz** and are mainly used for **short-range communication**. They cannot easily penetrate walls, which reduces interference between nearby rooms. However, this also prevents long-distance communication. Infrared is commonly used in **TV remote controls and short-range indoor communication**.

Medium	Main Feature	Examples
Twisted Pair	Low cost, widely used	Telephone, Ethernet
Coaxial	Higher bandwidth, good shielding	Cable TV, old Ethernet
Fiber Optic	Very high speed and bandwidth	Internet backbone
Radio	Omnidirectional	Radio, wireless
Microwave	Directional, line-of-sight	Cellular, satellite
Infrared	Short range, cannot penetrate walls	Remote control